

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION
NUMBER

BIOLOGY

9744/03

Paper 3: Long Structured and Free-response Questions

26 August 2022

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and Biology class in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your workings or if you do not use appropriate units.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/30
2	/10
3	/10
Section B	
4/5	/25
Total	/75

This document consists of **23** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

- 1** The influenza A virus subtype H7N9 has caused several epidemic waves since its appearance in 2013. H7N9 influenza virus causes high-mortality disease in humans. Monoclonal antibodies, made by cloning a unique white blood cell are being explored as treatments.

Fig. 1.1 shows how one such antibody, m826 antibody, could bind to hemagglutinin (HA) of the H7N9 influenza virus.

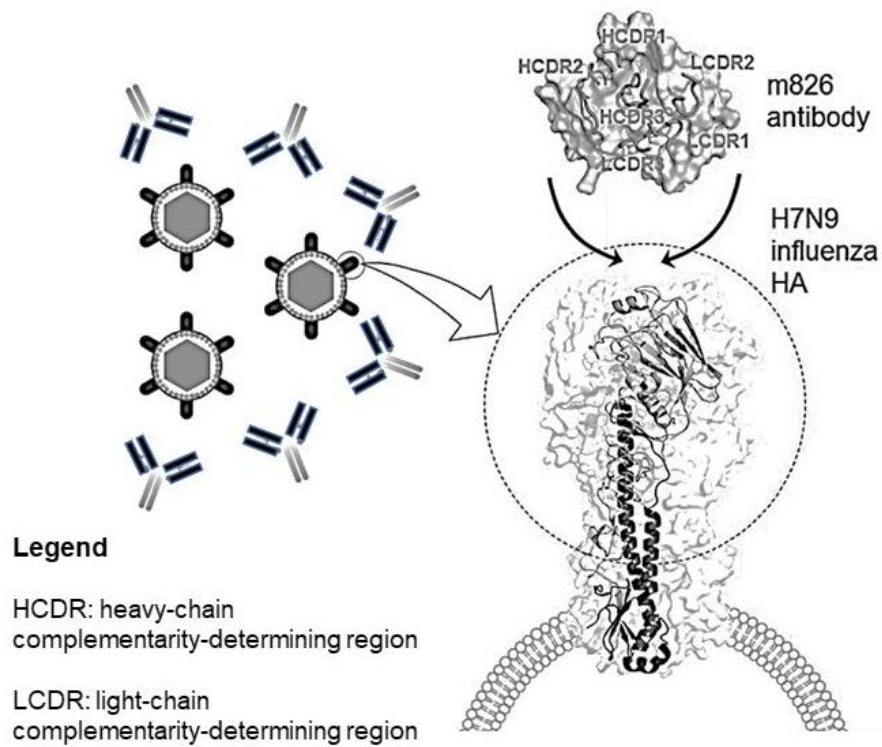


Fig. 1.1

- (a)** Name and describe the secondary structures of HA that are visible in Fig. 1.1.

[4]

- (b) With reference to Fig. 1.1, describe how m826 antibody could bind to HA of the H7N9 influenza virus.

[3]

- (c) Membrane glycoproteins such as HA consist of one or more short carbohydrate chains attached to the protein.

Outline the functions of such short carbohydrate chains on the cell surface membrane of the human epithelial cell.

[2]

- (d) HA is embedded in the influenza viral envelope that is derived from the cell surface membrane of a human epithelial cell of the respiratory tract.

State **two** structural similarities between the influenza viral envelope and the cell surface membrane of the human epithelial cell.

[2]

- (e) The fluid mosaic model of the cell surface membrane was elucidated with the aid of freeze-fracture electron microscopy. The steps involve freezing and fracturing the membrane before examining it with a scanning electron microscope, which produces images of an object's surface.

Fig. 1.2 shows the process of freeze-fracture electron microscopy.

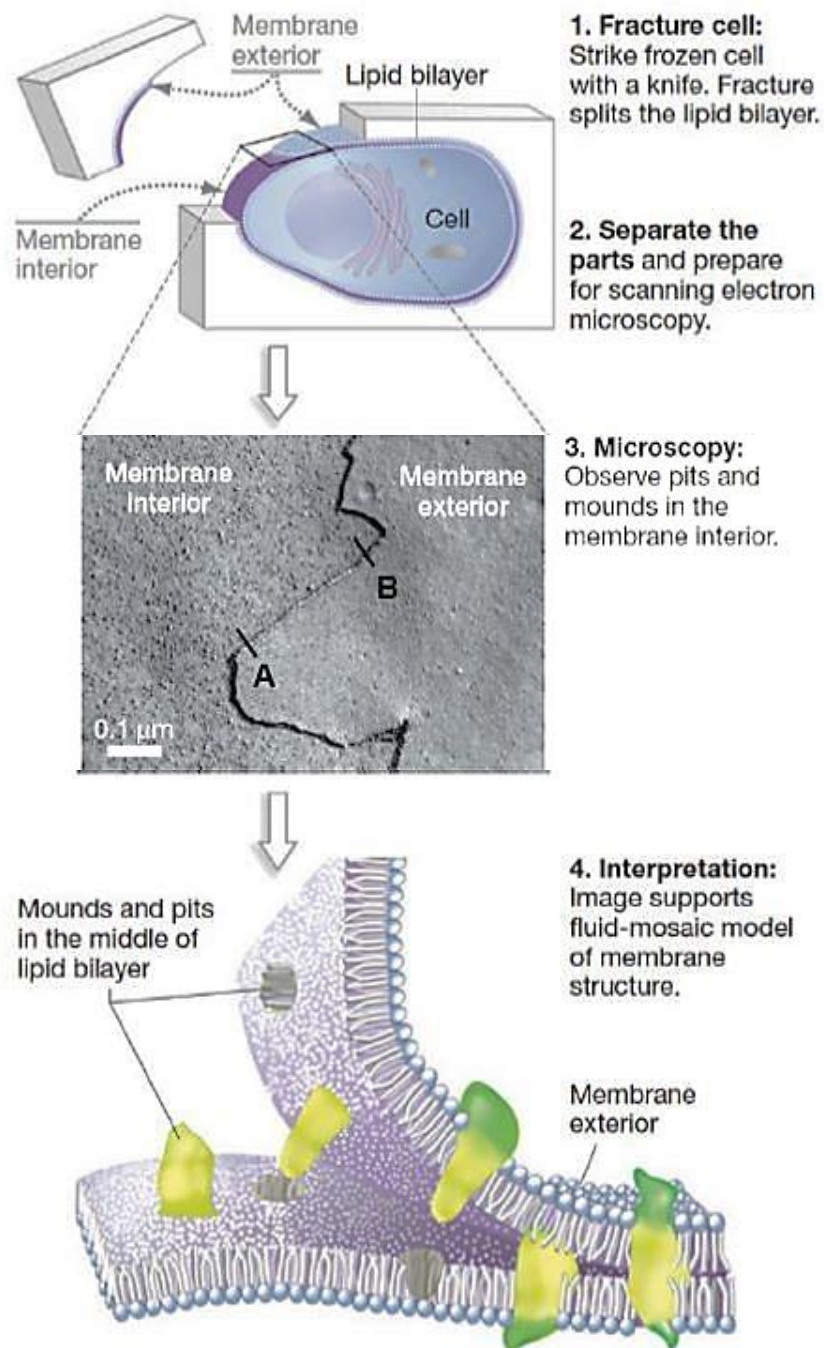


Fig. 1.2

- (i) Determine the distance between points **A** and **B** as shown in Fig. 1.2.
Show your working clearly in the space provided.

distance between points **A** and **B** = _____ μm [3]

- (ii) Suggest an appropriate control to show that the mounds and pits were not irregularities in the lipid bilayer caused by the freeze-fracture process.

[1]

- (f) Class switching results in millions of different antibody molecules. During class switching, the variable region generated during V(D)J recombination remains unchanged, but it becomes linked to another segment of the constant region of the heavy chain.

Fig. 1.3 shows how human genomic IgM-IgD region may undergo transcription in a mature B lymphocyte to produce mRNA coding for secreted or transmembrane form of antibody molecules (IgM or IgD). Exons encoding the rearranged VDJ region and the various domains of IgM and IgD are shown in boxes.

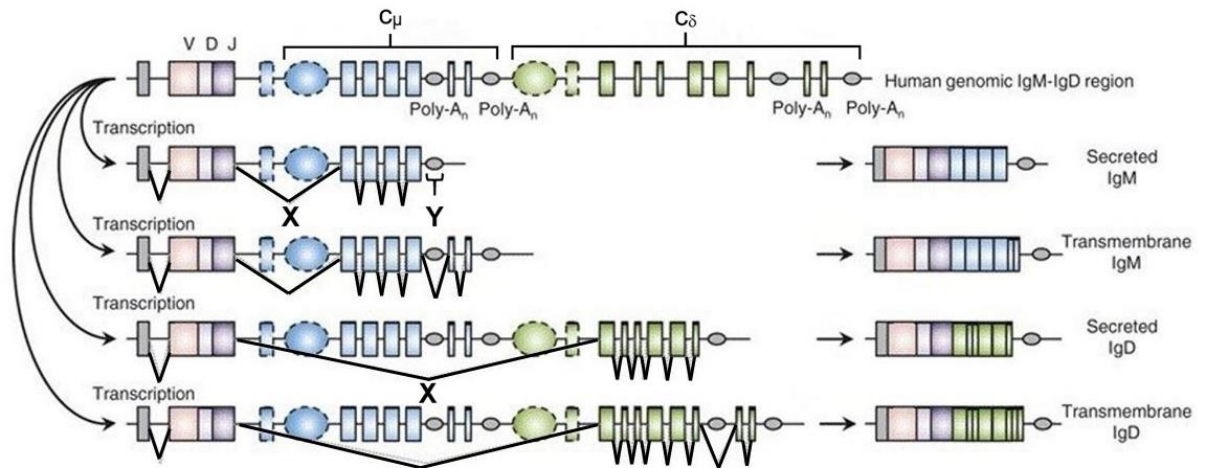


Fig. 1.3

- (i) State the **level** of control of gene expression as shown in Fig. 1.3.

[1]

- (ii) Name processes X in Fig. 1.3 and describe their significance.

[2]

- (iii) With reference to Fig. 1.3, identify region **Y** on the mRNA coding for secreted IgM and describe its roles.

[3]

- (iv) On Fig. 1.3, circle the region of the mRNA that codes for the transmembrane region of IgD.

[1]

- (v) Explain the structure and function of the transmembrane region of IgD.

[2]

- (g) Fig. 1.4 shows the excess deaths caused by pneumonia and influenza over a 50-year period in Country P and Country Q. The five-year moving average is also shown. The shaded bars show the rates of use of vaccine in each country.

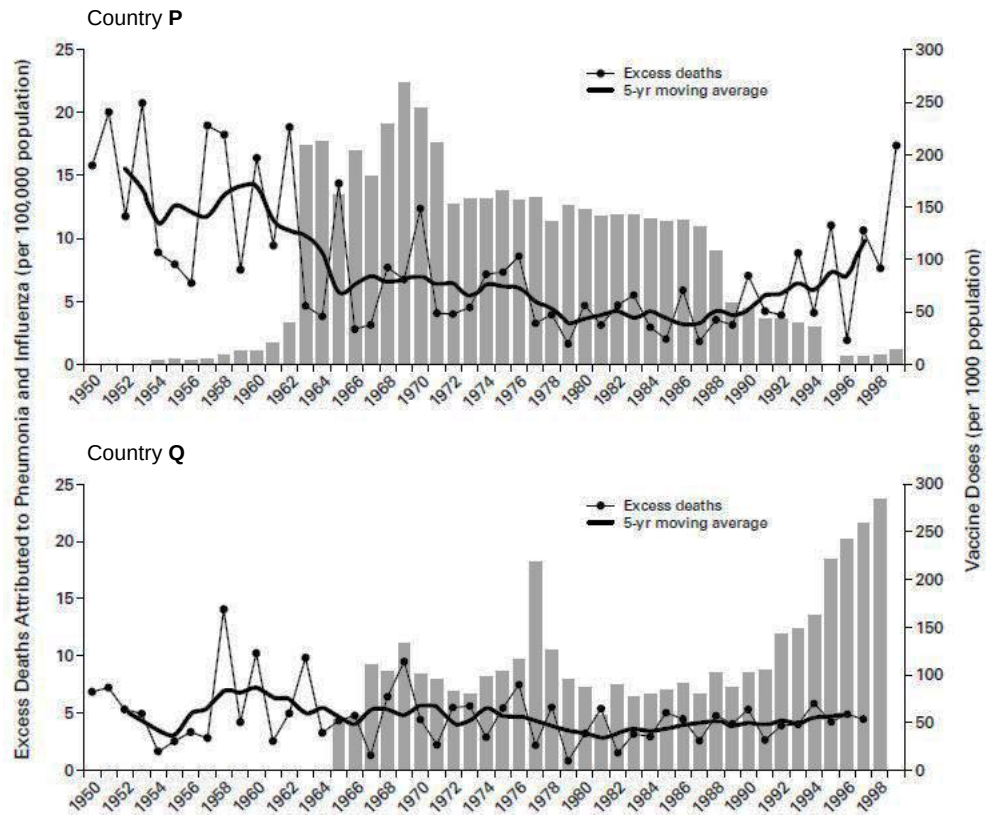


Fig. 1.4

- (i) With reference to Fig. 1.4, describe the effect of vaccination on excess deaths caused by pneumonia and influenza in the two countries.

[4]

- (ii) The policies for the control of pneumonia and influenza differed between Country **P** and Country **Q**. In Country **P**, vaccination was recommended for school children while in Country **Q**, vaccination was recommended for the elderly.

Using this information, suggest **one** reason for the trends described in (g)(i).

[2]

[Total: 30]

- 2 An increase in atmospheric carbon dioxide (CO₂) concentration and temperature affect the growth and yield of important crops such as potatoes.

In 2020, researchers investigated the effect of long-term exposure to elevated CO₂ concentration and temperature on potato plants.

The treatments involved the conditions: ambient CO₂ concentration and ambient temperature (aCaT), ambient CO₂ concentration and elevated temperature (aCeT), elevated CO₂ concentration and ambient temperature (eCaT), and concurrently elevated CO₂ concentration and temperature (eCeT). Conditions were maintained using a growth chamber for each of the treatments.

The ambient temperature was maintained at 20°C as modern potato cultivars generally grow well under moderate temperatures around 20°C while the elevated temperature was maintained at 4°C higher than ambient temperature.

Table 2.1 shows the details of the experiment conducted.

Table 2.1

	aCaT	eCaT	aCeT	eCeT
daily mean temperature / °C	20.9	20.9	24.9	24.9
daily mean CO ₂ concentration / ppm	417	771	425	804

- (a) Suggest why aCaT and aCeT use a CO₂ concentration at approximately 400 ppm.

[1]

Growth parameters such as plant height, leaf area and biomass were measured for six plants from each chamber at 58 days after emergence. Emergence is the growth of the plant towards the soil surface and coming out from the soil, making a shoot.

Fig. 2.1 shows the effect of different treatments on growth parameters at 58 days after emergence. Error bars are the lines above the vertical bars. Different letters above the error bars indicate significant differences between the treatments while same letters indicate no significant differences between the treatments.

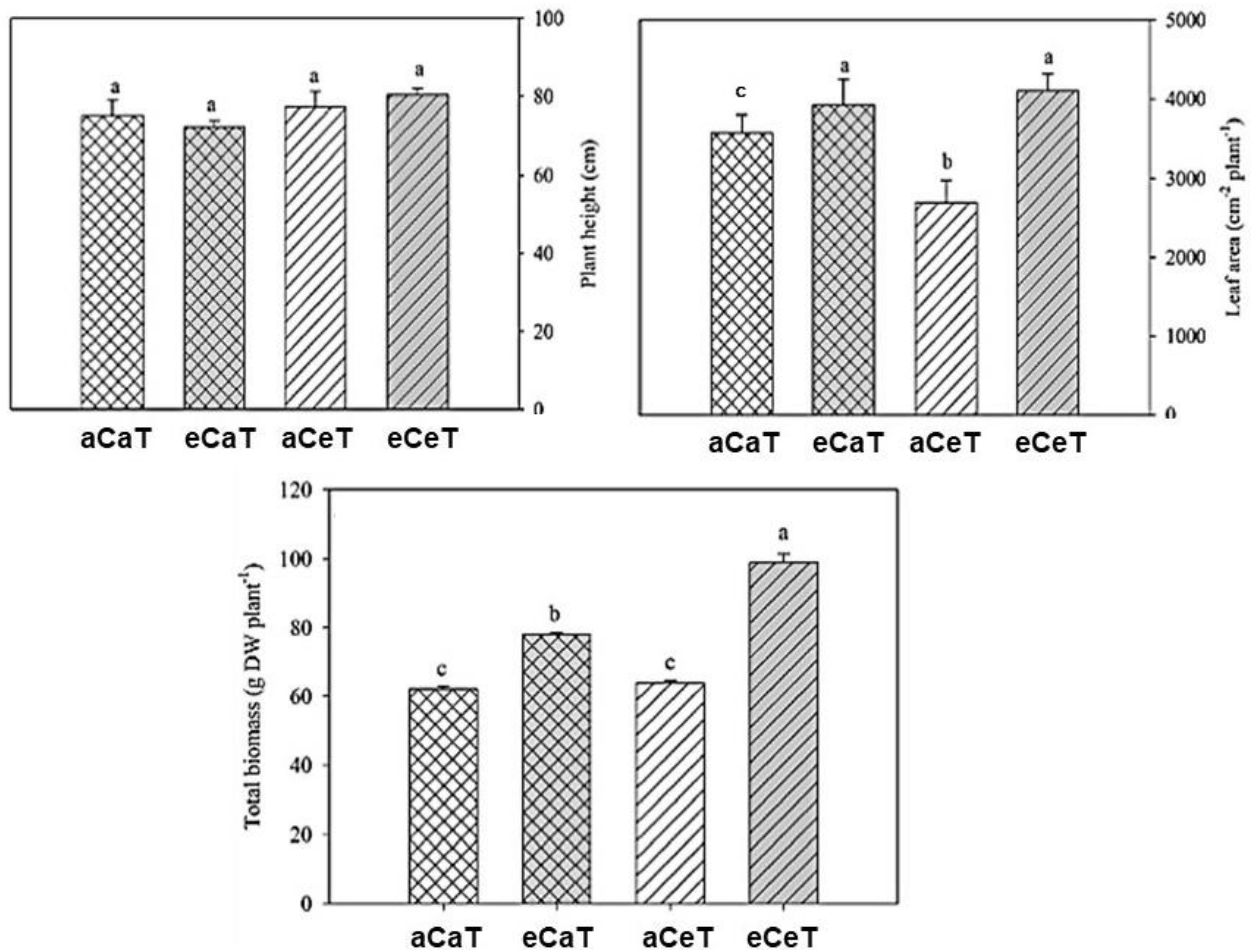


Fig. 2.1

- (b) Complete Table 2.2 to show the conclusions that could be drawn from Fig. 2.1.

Table 2.2

parameter	conclusion	
	elevated CO ₂ concentration	elevated temperature
plant height	no significant differences	
leaf area	significant increase	
total biomass		

[4]

- (c) With reference to Fig. 2.1 and your knowledge on photosynthesis, explain the results for total biomass in the treatment eCaT.

[2]

At harvest time, the average numbers of potato tubers and biomass of potatoes cultivated under varying CO₂ concentrations and temperatures were collected.

Table 2.3 shows the results.

Table 2.3

treatment	average numbers of potato tubers / plant ¹			biomass / g DW plant ¹
	< 30 g	30 – 80 g	> 80 g	
aCaT	4.8	2.9	2.3	106.4
eCaT	5.2	3.9	2.4	110.3
aCeT	2.0	2.8	1.9	76.9
eCeT	3.9	4.1	2.9	127.5

- (d) With reference to Table 2.3, describe the effect of elevated temperature on biomass of potato tubers at harvest time.

.....
 [1]

- (e) “Global warming due to increased atmospheric CO₂ concentration will lead to reduced global food supply”.

Discuss the validity of this statement.

.....

 [2]

[Total: 10]

- 3 The extreme coat-colour variations in oldfield mouse, *Peromyscus polionotus*, are driven by selection for camouflage. The mainland subspecies, *P. p. subgriseus*, has a coat that is dark brown on top and light grey on the belly, as well as a striped tail. The beach subspecies, *P. p. leucophalus*, has a light-coloured coat and lacks visible pigmentation on its face, sides and tail.

Fig. 3.1 shows the pigmentation of the two subspecies.

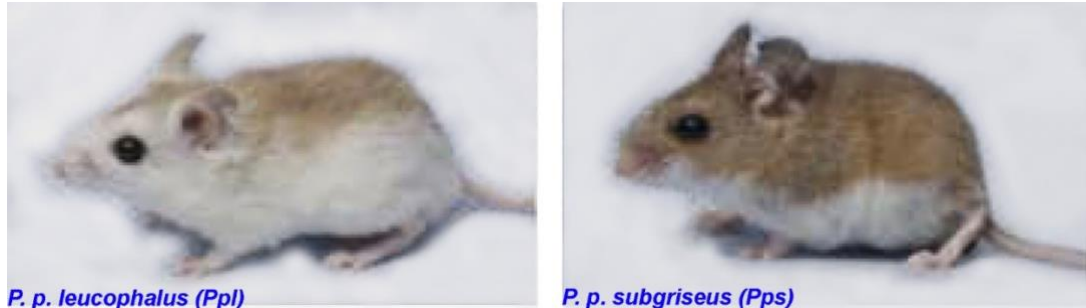


Fig. 3.1

Two genes, *melanocortin-1 receptor (Mc1r)* and its antagonist, *Agouti signaling protein (Agouti)*, located on independent chromosomes are responsible for most of the differences in pigmentation between the two subspecies. A receptor antagonist is a type of ligand that blocks or dampens a biological response when bound to the receptor.

To examine whether differences in the expression level of *Mc1r* or *Agouti* influence colour patterning, reverse transcriptase polymerase chain reaction [RT-PCR] was carried out on adult skin cells of the eyebrow area. The resultant RT-PCR product was analysed by gel electrophoresis.

Fig. 3.2 shows the results of the gel electrophoresis.

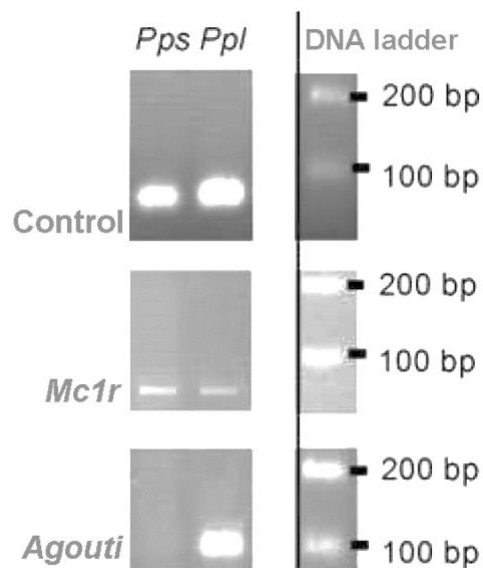


Fig. 3.2

(a) With reference to Fig. 3.2,

(i) describe the principle of gel electrophoresis.

[2]

(ii) describe the expression level of *Mc1r* and *Agouti*.

[2]

(b) RT-PCR is a laboratory technique combining reverse transcription of RNA into DNA and amplification of specific DNA targets using polymerase chain reaction. The technique is carried out on mRNA samples isolated from cells.

Suggest **one** advantage of RT-PCR compared to conventional PCR.

[1]

To analyse the genetic basis of pigmentation pattern differences, mainland mice were crossed with beach mice, yielding F_1 hybrids that were crossed with each other to produce F_2 progeny.

(c) Alleles for *Mc1r* and *Agouti* are represented by **D** for dark and **L** for light.

Table 3.1 shows the distribution of genotypes in 30 F_2 progeny lacking pigmentation on the rear end.

Table 3.1

	DD	DL	LL
<i>Mc1r</i>	0	0	30
<i>Agouti</i>	0	15	15

State the dominant allele for each of the following genes.

Mc1r

Agouti [2]

- (d) Fig. 3.3 shows the pigmentation in hairs for pigmented cheek area in 454 F₂ progeny clustered by *Agouti* genotype.

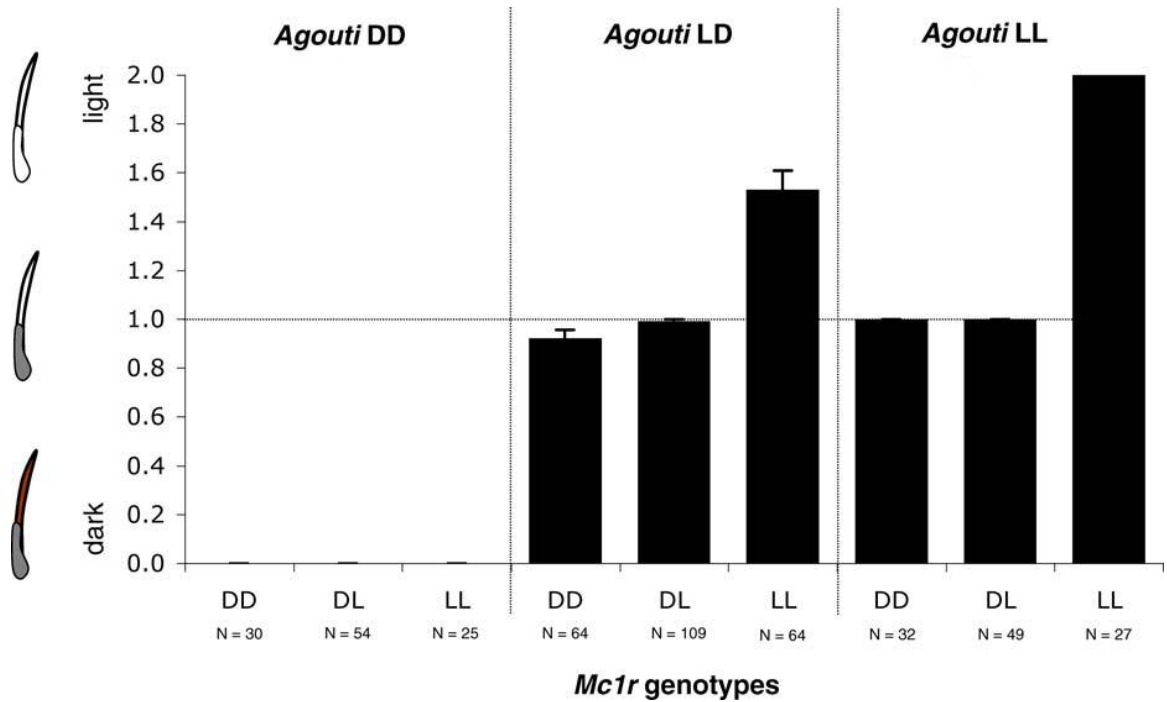


Fig. 3.3

The phenotype for each of 454 F₂ progeny was scored as 2 (no visible pigment), 1 (partially pigmented), and 0 (fully pigmented) as indicated by the cartoons of individual hairs. N refers to the number of individuals in each category.

Name and describe the type of gene interaction as shown in Fig. 3.3.

[3]

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

- 4 (a)** Describe how a functional enzyme is synthesised in bacteria. [15]
- (b)** Outline the process of transduction and its advantages to the bacteria. [10]
- [Total: 25]
-
- 5 (a)** Describe the structure-function relationship of the endomembrane system in plants. [15]
- (b)** Outline the process of photophosphorylation and its advantages. [10]
- [Total: 25]

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